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Cognitive-Behavioral Family Therapy for Anxiety-Disordered Children: A Multiple-Baseline Evaluation[1]

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Six children (aged 9 to 13) diagnosed with a childhood anxiety disorder were treated with an 18-session, family-based cognitive-behavioral therapy that was evaluated using assessments from multiple sources and a multiple-baseline (2, 4, and 6 weeks) across-cases design. Diagnoses, parent and teacher reports, and child self-reports assessed outcome. Changes in diagnostic status, standardized parent- and teacher-report measures, and parent and child reports on specific measures of coping indicated meaningful treatment-related gains. Gains were considered clinically significant and were, in general, maintained at 4-month follow-up. Although measured features of the family were of limited use, the results suggested the utility of a family-based treatment for childhood anxiety disorders.

KEY WORDS: childhood anxiety; anxiety disorders; cognitive-behavioral therapy; child clinical; family therapy; treatment outcome.

Investigations of anxiety and fear in children have played a pivotal role in the development of theories of personality and psychopathology and in the development of interventions such as modeling (Bandura, Grusec, & Menlove, 1967) and desensitization (Wolpe, 1958). Treatment of child anxiety disorders has more than historical value, however. Internalizing problems of anxiety, social withdrawal, hypersensitivity, and depression have been

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found in 10% to 20% of school-aged children (Orvaschel & Weissman, 1986; Werry, 1986; see also Anderson, 1994). These internalizing problems affect passage through the normal developmental challenges of childhood and adolescence (Kendall et al., 1992). Studies have shown the negative impact of anxiety on academic success (e.g., Dweck & Wortman, 1982) and social adjustment (e.g., Strauss, Lease, Kazdin, Dulcan, & Last, 1989), with implications for adult functioning. For example, a significant number of adults with anxiety disorders reported that they suffered from anxiety symptoms during childhood (Last, Hersen, Kazdin, Francis, & Grubb, 1987; Last, Phillips, & Statfeld, 1987; Weissman, Leckman, Merikangas, Gammon, & Prusoff, 1984).

Given the impact on child adjustment and the relationship between child anxiety disorders and adult functioning, development of effective interventions in treating child anxiety is essential. Kane and Kendall (1989) and Kendall (1994) reported favorable results of a cognitive-behavioral manualized treatment (Kendall, Kane, Howard, & Siqueland, 1990) that was child-focused. Results indicated that approximately two-thirds of treated subjects no longer had their primary diagnosis as primary at posttreatment and at 1-year and 3.35-year (Kendall & Southam-Gerow, 1996) follow-ups.

The critical importance of family context in understanding and intervening with anxious children has been increasingly recognized (e.g., Kendall, MacDonald, & Treadwell, 1995). One approach to address the importance of family involvement in treatment is parent training (Fauber & Long, 1991). To date several case studies have included parent involvement in a behavioral or cognitive-behavioral intervention for an anxiety-related disorder (e.g., Hagopian, Weist, & Ollendick, 1990; Mansdorf & Lukens, 1987). A study conducted in Australia evaluated a shortened version of the child-focused intervention developed and reported by Kendall (1994) to the same shortened treatment with the addition of a parent-training component (Barrett, Dadds, & Rapee, 1996). It was found that the inclusion of parent training resulted in some enhanced outcomes on some measures. The present authors developed a manualized 18-session family-based cognitive-behavioral therapy (Howard & Kendall, 1992) as an extension of our earlier efforts to involve parents. Some studies have suggested that children at greatest risk for anxiety disorders come from family environments characterized by rigid organization and control, those providing less independence for children, and those with decreased tolerance for expression of anger and other negative emotions (e.g., Siqueland, Kendall, & Steinberg, 1996). The manualized family-based intervention evaluated in this study remains child-focused, but goes beyond parent training in making the family part of the solution in treating child anxiety disorders. Other family members are actively involved in looking at their own experience with anxiety and their attempts to cope with anxiety-provoking situations, including parental anxiety about supporting the child's developing autonomy.

The present study is a multiple-baseline evaluation of the effectiveness of a family-based treatment. Assessment of treatment effects uses child-, parent-, and teacher-report measures and diagnoses. The participating children and their families were expected to show improvement on measures assessing internalizing symptoms and anxious behavior following treatment, but not to show improvement on the same measures during varying durations of baseline. Following intervention, but not following baseline, subjects were expected to no longer qualify for an anxiety disorder diagnosis. Treatment effects were expected to be maintained during a 4-month follow-up period. In addition, family report measures were included as a very preliminary examination of an association between family style or marital conflict and child anxiety disorder diagnosis.

METHOD

Participants

Six children who met *Diagnostic Statistical Manual of Mental Disorders* (3rd ed., rev.) (DSM-III-R; American Psychiatric Association, 1987) diagnostic criteria for an anxiety disorder and their families completed treatment. Cases were consecutive referrals to the Child and Adolescent Anxiety Disorders Clinic (CAADC) at Temple University. Exclusion criteria included a primary diagnosis of a specific phobia, a disabling physical condition, psychosis, or current use of antianxiety medications.

Initial diagnoses, and comorbid diagnoses, are indexed in Table I. Participants were three 9-year-old boys (third grade), one 10-year-old boy (fourth grade), one 13-year-old male adolescent (eighth grade), and one 11-year-old girl (sixth grade). At the start, all families were intact two-parent families; the parents of Subject 6 separated during treatment.

Measures

Structured Diagnostic Interviews

Anxiety Disorders Interview Schedule for Children (ADIS-C; Silverman & Nelles, 1987). This structured diagnostic interview has a compatible parent form, was designed specifically for the diagnosis of childhood anxiety

Table I. Diagnoses and Severity Ratings over Time[a]

Subject	Diagnostic source	Baseline Anxiety	Baseline Other	Pretreatment Anxiety	Pretreatment Other	Posttreatment Anxiety	Posttreatment Other	Follow-up Anxiety	Follow-up Other
1	Parent	OAD-4	Sch-Phob-4 ODD-4	OAD-4	Sch-Phob-4 ODD-4	—	ODD-2	—	ODD-2
	Child	SAD-2	Sch-Prob-4	SAD-2	Sch-Phob-4	SAD-2	—	—	—
2	Parent	SAD-2		SAD-2		—		—	
	Child			SAD-2		—		—	
3	Parent	OAD-4		OAD-3		Avoidant-4		—	Social Phob-3
	Child	OAD-2 SAD-4		OAD-2 SAD-4		— —		— —	
4	Parent	OAD-4	ADHD-3	OAD-4	ADHD-3	—	—	—	—
	Child	OAD-4	Simple Phob-4	OAD-4	Simple Phob-4	—	—	OAD-2	—
5	Parent	OAD-3	ADHD-2	OAD-3	ADHD-3 Simple Phob-4	OAD-2	ADHD-3 Simple Phob-3	—	ADHD-3 —
	Child	OAD-3	ADHD-2	OAD-3	ADHD-2	—	ADHD-2	—	—
6	Parent	OAD-2		OAD-3		—	ODD-4	OAD-2	Dysthymia-3
	Child	OAD-2		OAD-2		—		OAD-2	Simple Phob-3 Depression-3

[a]OAD = overanxious disorder; School Phob = school phobia; ODD = oppositional defiant disorder; SAD = separation anxiety disorder; ADHD = attention deficit hyperactivity disorder; Simple Phob = simple phobia. Dashes indicate the later absence of a pretreatment diagnosis. Subject 1: 10-year-old male; 2: 9-year-old male; 3: 13-year-old male; 4: 11-year-old female; 5: 9-year-old male; and 6: 9-year-old male.

disorders, and allows the diagnosis of other DSM disorders. Reliability and validity with other structured interviews and self-report measures has been reported to be adequate.

Child Self-Report Measures

Fear Survey Schedule for Children—Revised (FSSC-R; Ollendick, 1978). The FSSC-R assesses specific fears in children and has shown good internal consistency and retest reliability, and normative data are available (Ollendick, 1983). Studies have shown the FSSC-R to be responsive to intervention, but mixed results in discriminating treated and untreated groups (Johnson & Melamed, 1979).

Revised Children's Manifest Anxiety Scale (RCMAS; Reynolds & Richmond, 1978). The RCMAS measures a child's chronic or trait anxiety. The 37-item scale includes an 11-item Lie scale. The RCMAS has demonstrated validity and national reliability and normative data are available (Reynolds & Paget, 1983). The RCMAS has been shown to differentiate children with anxiety disorders from control groups with other psychiatric diagnoses and from nonanxious controls (Bell-Dolan, Last, & Strauss, 1990).

Coping Questionnaire—Child (CQ-C). This instrument (Kendall, 1994) assesses children's self-perceived ability to cope with specific anxiety-provoking situations. Three situations identified by a child and parent as most severe at the diagnostic interview are listed on the Coping Questionnaire. The child's perception of ability to cope with each situation is rated on a 1- to 7-point scale, with the average of the three ratings used to measure changes in the child's self-perceived coping.

The State-Trait Anxiety Inventory for Children (STAIC; Spielberger, 1973). This instrument includes a 20-item scale to measure enduring tendencies to experience anxiety (A-Trait) and a 20-item scale to measure temporary variations in anxiety (A-State). Normative data are available. Concurrent validity has been shown in correlations with behavioral observations and other anxiety measures (Kendall & Ronan, 1990; Spielberger, 1973).

Children's Depression Inventory (CDI; Kovacs, 1981). This 27-item measure assesses the cognitive, affective, and behavioral symptoms of depression. Children select which of the three choices for each item best describes them during the preceding 2 weeks. Standardization data are available for children from 7 to 17 years. The CDI correlates with ratings of depression and with DSM-III depression diagnoses (Kendall, Cantwell, & Kazdin, 1989; Saylor, Finch, Shapiro, & Bennet, 1984).

Parent Report Measures

Child Behavior Checklist—Parent Form (CBCL; Achenbach & Edelbrock, 1983). The 118-item (scored 0 to 2) CBCL assesses child behavioral problems and social competencies and provides scores for internalizing and externalizing problems. The CBCL is widely used and has reliability, validity, and extensive normative data for both normal and clinical populations (Achenbach & Edelbrock, 1983).

Coping Questionnaire—Parent (CQ-P). Like the CQ-C this scale measures children's ability to cope with three specific anxiety-provoking situations identified during the diagnostic interview (Kendall, 1994). Parent(s) use a 1- to 7-point scale to rate child coping.

State-Trait Anxiety Inventory for Children—Modification of Trait Version for Parents (A-Trait-P; Strauss, 1987). This measure is a modification of the trait version of Spielberger's (1973) STAIC to be used for parent report of child anxiety.

Family Self-Report

Self-Report Family Inventory (SFI; Beavers, Hampson, & Hulgus, 1985). The 36-item SFI was designed to evaluate each family member's perception of their family functioning. Each member (9 years of age or older) individually answers items to assess family health/competence, conflict, communications, cohesion, directive leadership, and expressiveness. These factors are congruent with those obtained from the Beavers observational ratings scales. Normative data are available and adequate internal consistency and retest reliability for most factors has been reported (Beavers, Hampson, & Hulgus, 1990).

O'Leary-Porter Scale (OPS; Porter & O'Leary, 1980). The OPS is a 10-item scale developed to assess the frequency of overt parental conflict that occurs in front of the child. The items are rated by parents (5-point Likert-type scale) and total scores range from 0 to 40. A 2-week retest reliability of .96 and concurrent validity based on responses to other marital conflict measures have been reported (Porter & O'Leary, 1980). For this research, a Modified O'Leary-Porter Scale was used. An additional 10 items with the same content, but measuring *covert* (discreet) conflict—parental conflict that does not take place in front of the children—were included.

Teacher Report

Child Behavior Checklist—Teacher Report Form (TRF; Achenbach & Edelbrock, 1986). The TRF, using 0- to 2-point ratings by the primary teacher, provided data on each child's classroom functioning and anxious behavior at school. The TRF is widely used and has reliability, validity, and extensive norms for both normal and clinical populations (Achenbach & Edelbrock, 1983).

Procedure

Following referral, an intake evaluation by a trained diagnostician was scheduled. Parents and child signed informed consent forms and completed the structured interview and questionnaire measures. Subjects were randomly assigned to a baseline duration prior to the family-based cognitive-behavioral treatment.

Treatment was initiated after baseline assessment periods of 2, 4, or 6 weeks to produce a multiple-baseline design (Barlow & Hersen, 1984; Kazdin, 1986). Maintenance of treatment effects was assessed during and at the end of 4-month follow-up.

The structured diagnostic interview was administered prebaseline, at pre- and posttreatment and at follow-up. Pencil-and-paper measures were gathered at prebaseline, at pretreatment, during treatment, at posttreatment, during follow-up, and at 4-month follow-up. Prebaseline, pre-, mid-, and posttreatment, and follow-up assessments included parent reports, and prebaseline, pre-, and posttreatment, and follow-ups included teacher reports. In addition, during baseline, during treatment, and during follow-up, weekly assessments were obtained from the child using the CDI, the RCMAS, and the CQ-C. Weekly parent reports of the child's symptoms and coping were also obtained.

Setting and Therapist

Therapy, provided by the first author, an experienced senior doctoral candidate, was conducted at the CAADC.

Intervention

Families received an average of 18 (16 to 20) 50-min sessions. All family members attended the first two sessions, and later sessions included at least one family member in addition to the target child. A portion of every session included at least one parent. Sessions were typically held once a week, with variations for holidays, illness, and schedule conflicts.

Therapy followed a 97-page manual (Howard & Kendall, 1992) that outlines a cognitive-behavioral family treatment for child anxiety disorders. The manual describes the goals and strategies for each session, but tasks and application were individualized to be clinically sensitive to each child's level of cognitive development, as well as the child's specific anxiety-provoking situations, family structure, and family capacity to help the child master new situations.

The manual uses cognitive-behavioral strategies (see also *Coping Cat Workbook*, Kendall, 1990), including (a) recognizing anxious feelings and somatic reactions, (b) identifying anxious cognitions in anxiety-provoking situations, (c) developing problem-solving steps to cope with the anxious reactions, and (d) self-evaluating performance and self-reward. The training uses behavioral strategies such as relaxation training, modeling, role play, in vivo exposure, and contingent reinforcement, particularly social reinforcement, to build the above coping skills. There is a graduated sequence of learning tasks and assignments, and "Show-That-I-Can" (STIC) tasks are assigned to be completed between sessions.

The family-based treatment emphasizes (a) initial observation of family interaction to assess the level of family functioning and family style; (b) therapist structuring of session interactions to encourage expression of negative affect and differences, problem negotiation, and respect for the child's opinions and experience; (c) using therapist questions and observations about the presenting problem to normalize the experience of anxiety and to encourage understanding of the child's symptoms in the wider context of how relationships within the family and between the family and other systems are negotiated; and (d) family participation in the sequence of learning tasks and assignments, with the form of participation varied to fit the family's hypothesized role in development or continuation of the child's presenting problems.

The treatment was divided into two segments. The first half was devoted to exploring the interpersonal nature of each child's difficulties, increasing flexibility in thinking about the problem, and learning active strategies to change the child and family members' usual way of reacting. The second half was for practice, with a focus on the application of skills in situations where the child had shown anxious behavior. The practice situations served to structure family members' involvement, working directly with problematic parental responses to the child's experiencing distress and entering unfamiliar situations. The specific situations were individually designed, based on his/her own particular distress as assessed during initial sessions. Practice was graduated, beginning with imaginal, in-office exposure to relatively nonstressful situations and building to situations in which the child was typically extremely anxious, unable to cope, and distressed.

Depending on family members' involvement in coaching use of new skills, between-sessions practice with one or more family members followed the successful completion of each level during a therapy session. Parents and other family members were invited to select situations where they would like to act more effectively and to themselves apply the treatment strategies as a way of helping the child learn. When parents were overprotective and experienced excessive anxiety about the child's experiencing discomfort or becoming more autonomous, this was a direct focus of therapy—with the therapist helping explore parental assumptions and characteristic reactions and their consequences.

Treatment Integrity

An assessment of treatment integrity was done by a CAADC staff member trained in the provision of cognitive-behavioral treatment (therapist in National Institute of Mental Health-funded outcome study). A checklist adopted from this earlier study (Kendall, 1994) was used. The rater completed the checklist by watching 35 randomly selected 3-min segments of randomly selected videotaped treatment sessions (including each subject).

RESULTS

Diagnostic Reliabilities

Diagnostic interviewers reached 85% agreement during training. Ongoing diagnostic reliabilities (kappa; Cohen, 1977) using randomly selected interviews found 100% interrater reliability for the anxiety disorders.

Treatment Integrity

The integrity check showed that the segments that were reviewed were consistent with the treatment as described in the manual. The treatment strategies outlined in the manual were applied in all cases.[3]

[3]Subject 6 participated in a school-based group for children in families where parents were separated or divorced and attended one family evaluation session at another clinic before the end of the follow-up period. There were no other instances where other forms of therapeutic intervention were used during the baseline, treatment, or follow-up phases.

Treatment Outcomes

Diagnoses. Cases, based on ADIS-C-P, received a primary child anxiety disorder diagnosis at intake and continued to meet diagnostic criteria at the end of the 2-, 4-, or 6-week baseline (see Table I). In contrast, therapy was linked to change in anxiety diagnoses. Table I shows that diagnosticians, who were blind to subjects' treatment status, determined that four of six cases had no child anxiety diagnosis at posttreatment. Using the child interview (ADIS-C-C), Table I shows that, after treatment, five of six cases received no anxiety diagnosis.

Parent Report. Parent-reported internalizing problems (CBCL *T*-scores) showed improvement during treatment, but not over the baseline period. Mean baseline, pretreatment, posttreatment, and follow-up Internalizing *T*-scores, as well as scores for each child, are shown in Table II. Five subjects' *T*-scores began above a clinical cutoff of 70 (at both baseline and pretreatment) and all six subjects' scores were reduced to below the cutoff at posttreatment and follow-up. A similar pattern of change was evident on the STAIC-T-P.

CQ-P. Weekly parent reports of the child's ability to cope showed significant improvement with treatment. Figure 1 shows that CQ-P scores were relatively stable over baseline and increased with treatment. Visual inspection shows change (improvement) related to treatment for each child.

Teacher Report. Teacher-reported internalizing problems (TRF *T*-scores) showed gains for all children following treatment but no meaningful change during baseline. *T*-scores are shown for each subject (and means) in Table II. Five of six subjects had clinically significant internalizing symptoms (*T*-scores > 70) at the beginning and end of baseline, but not following the family-based therapy. Note that teachers were not a part of treatment, that the measures were solicited by the CAADC using procedures to maintain the teachers being blind to each child's treatment status. It cannot be ruled out, however, that information about the treatment was shared by parents or children.

Child Report. Mean FSSC scores are shown in Table III and suggest a meaningful reduction in the number of fears following treatment, although no such change was seen during baseline. In addition, five of the six individual subjects showed a marked reduction in the number of fears with treatment (one subject's scores remained essentially unchanged).

Mean STAIC Trait scores for the combined subjects are also shown in Table III. In addition, individual baseline, pretreatment, posttreatment, and follow-up scores are shown. Combined scores show substantial gains over treatment, but not over baseline. Further, five of six children self-reported reductions in trait anxiety following treatment.

Table II. Parent and Teacher Scores on Measures of Internalizing Symptoms[a]

Scale and source	Subject	Assessment points: Baseline	Pre-treatment	Mid-treatment	Post-treatment	4-Month follow-up
CBCL						
Parent	1	77	79	77	57	63
(Int. *T*)	2	68	73	68	49	48
	3	71	73	67	62	58
	4	74	75	76	53	40
	5	70	63	58	57	58
	6	71	73	67	64	69
	Mean	71.8	72.7	68.8	57.0	56.0
TRF						
Teacher	1	75	73	NS	62	NO
(Int. *T*)	2	92	89	NS	64	62
	3	51	62	NS	68	76
	4	72	73	NS	58	NO
	5	92	82	NS	64	NO
	6	71	73	NS	61	NO
	Mean	75.5	75.3		62.8	69.0
STAIC-T-P						
Parent	1	54	61	50	38	41
	2	44	50	38	28	28
	3	56	55	43	37	28
	4	62	57	48	32	37
	5	52	46	40	38	38
	6	43	46	49	33	41
	Mean	51.8	52.5	44.67	34.3	35.5

[a]CBCL (Int. *T*) = Child Behavior Checklist (Internalizing *T*-score); TRF (Int. *T*) = Teacher Report Form (Internalizing *T*-score); STAIC-T-P = the parent version of the trait portion of the State-Trait Anxiety Inventory for Children. NO = Not obtained; NS = Not sought.

Changes in trait anxiety (weekly RCMAS) evidenced some variability and some initially low scores. Nevertheless, the scores of four of the five subjects who had at least one *T*-score of 60 or above during baseline decreased to scores consistently below 60 by the end of treatment.

CDI. Four of six children self-reported depressive symptoms (CDI > 13) at pretreatment. By the end of treatment, CDI scores for each were below a clinical level, providing suggestive evidence that decreases in depressive symptoms paralleled decreases in self-reported anxiety.

CQ-C. Similar to parent reports, child weekly coping scores (shown in Fig. 2) evidenced a relatively stable baseline. Five subjects whose coping scores (averaged across their three target complaints) were problematic at the beginning showed meaningful improvement over treatment.

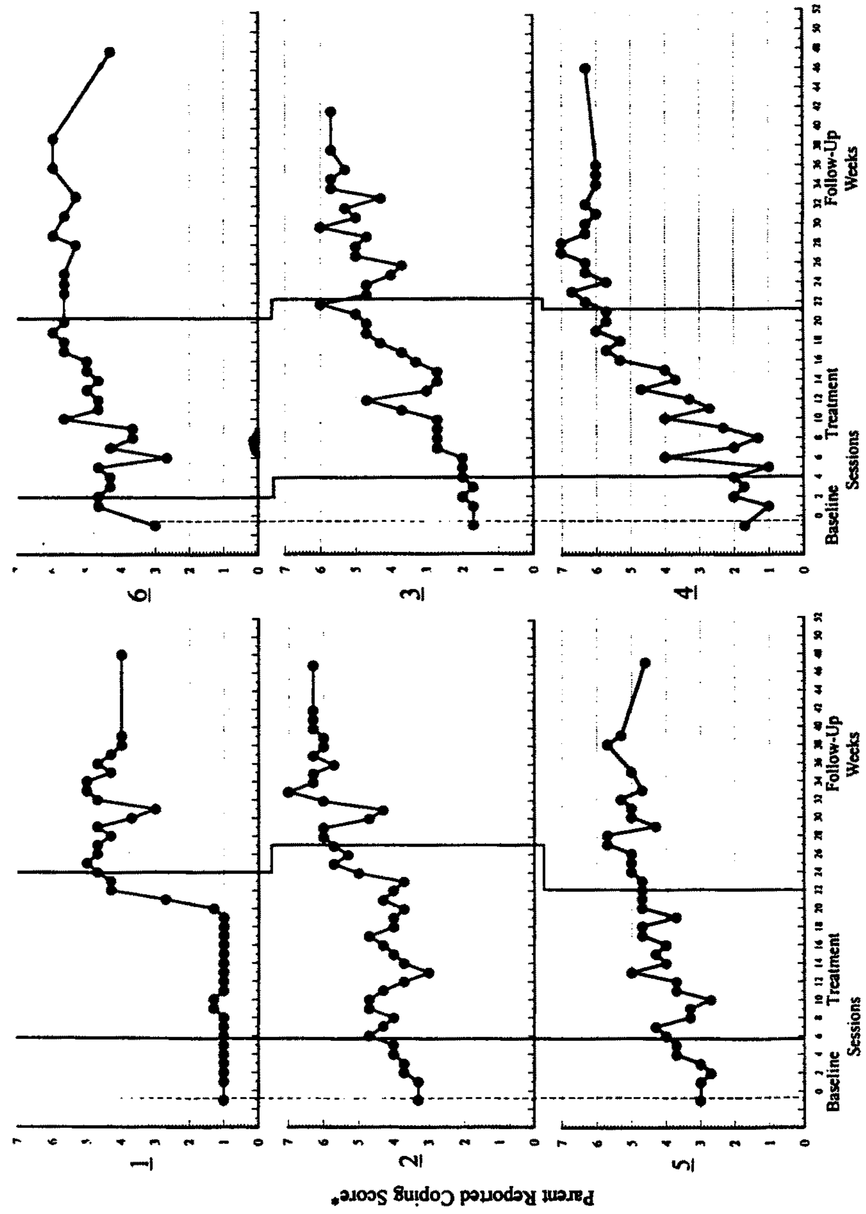

Fig. 1. Changes in parent-reported child coping skills. [*Average of scores for three target complaints for six subjects across baseline and treatment (sessions) and across weeks during follow-up.]

Table III. Scores on Child Self-Report Measures[a]

Measure	Subject	Baseline	Pre-treatment	Mid-treatment	Post-treatment	4-Month follow-up
		Assessment points				
FSSC-R	1	148	159	149	134	116
	2	116	100	88	88	80
	3	127	135	118	91	87
	4	187	166	169	160	132
	5	117	119	128	119	126
	6	160	148	145	132	113
	Mean	142.5	137.8	132.8	120.7	109.0
STAIC-T	1	67	47	57	42	56
	2	31	22	31	22	25
	3	60	52	38	38	36
	4	74	65	53	34	32
	5	60	60	41	41	29
	6	53	42	51	55	46
	Mean	57.5	48.0	45.2	38.7	37.3
CDI	1	16	15	10	8	9
	2	0	1	0	1	0
	3	13	10	3	3	2
	4	11	17	9	4	10
	5	42	24	14	12	10
	6	27	20	18	8	26
	Mean	18.16	14.5	9.0	6.0	9.5

[a]FSSC-R = Fear Survey Schedule for Children—Revised; STAIC-T = State-Trait Anxiety Inventory for Children—Trait; CDI = Child Depression Inventory.

Family Measures. Parental reports of the level of covert (discreet) or overt marital conflict using a modified OPS are shown in Table IV. The scores do not suggest a pattern of changes in parental conflict associated with this treatment.

The SFI data were plotted on the Diagram of Family Assessment (Beavers et al., 1990). The results for each family were examined, but are not presented here. Essentially, there were no meaningful patterns in the results: The data did not suggest that family functioning and style in a family with an anxiety-disordered child reflected a particular family style or specific family dysfunction.

Maintenance of Change

In general, treatment gains were maintained during the 4-month follow-up period for five of the six subjects. At the end of the 4-month fol-

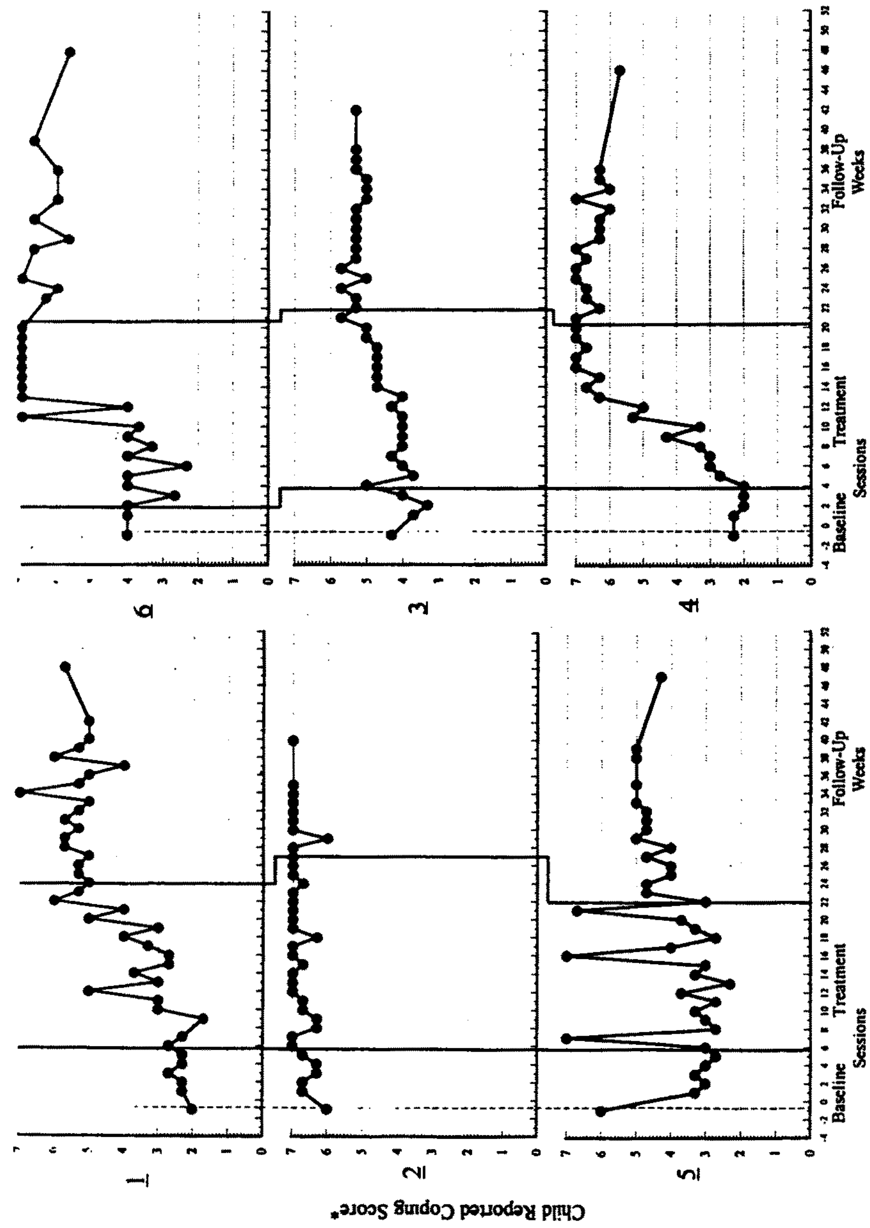

Fig. 2. Changes in child-reported coping skills. [*Average of scores for three target complaints for six subjects across baseline and treatment (sessions) and across weeks during follow-up.]

Table IV. Overt and Covert Conflict Scores for the Modified O'Leary-Porter Scale, Over the Assessment Points of Treatment

		Assessment points		
Subject	Source	Pretreatment	Posttreatment	Follow-up
		Overt conflict		
1	Mother	14	10	11
	Father	11	11	11
2	Mother	18	14	15
	Father	15	14	15
3	Mother	14	10	11
	Father	17	16	17
4	Mother	18	16	13
	Father	16	15	14
5	Mother	13	14	14
	Father	5	13	13
6	Mother	23	26	27
	Father	22	27	NO[a]
	Means	15.50	15.50	14.63
		Covert conflict		
1	Mother	12	11	11
	Father	11	10	11
2	Mother	18	13	14
	Father	9	8	9
3	Mother	13	8	10
	Father	16	15	15
4	Mother	17	14	10
	Father	16	12	14
5	Mother	8	9	10
	Father	6	10	31
6	Mother	23	25	31
	Father	22	27	NO[a]
	Means	14.25	13.50	13.09

[a]NO = Not obtained.

low-up period, five of six children received no child anxiety disorder diagnosis using the parent structured clinical interview and four of six received none using the child interview (see Table I). At 4-month follow-up, CBCL scores remained below a clinical cutoff, and all five subjects with clinically significant Internalizing T-scores prior to intervention maintained scores within normal range at the end of follow-up (see Table II). Two of three subjects for whom follow-up TRF scores are available maintained treatment gains during follow-up. The third, an adolescent who became more vocal about his concerns, actually received a T-score $>$ 70 for the first time at follow-up. Both child- and parent-report measures of anxious and depressive symptoms also showed maintenance.

There were two interesting additional outcomes. First, on many measures for some cases, treatment-related gains were enhanced during the 4-month follow-up period, a result that was not anticipated. Involving parents and siblings in a family-based cognitive-behavioral child-focused therapy may enhance treatment gains as parents continue to support their child's improved coping and greater autonomy after treatment ends. Second, Subject 6 initially showed treatment-related gains on all measures, but serious family stress near the end of the treatment period—including marital separation and domestic violence—resulted in mixed symptoms of anxiety and depression during the follow-up interval.

Clinical Significance

Meaningful assessment of treatment outcome requires consideration of the clinical significance of treatment-related change. Clinically significant improvement can be defined as gains that return deviant subjects to within normal limits on measures for which normative information is available (Kendall & Grove, 1988). The present research showed clinically significant change. First, following treatment, six of six CBCL Internalizing T-scores, six of six TRF Internalizing T-scores, six of six RCMAS T-scores, six of six CDI scores, and six of six STAIC T-scores were below a clinically significant level (T-score $<$ 70). This finding was supported by the fact that, following treatment, four of six subjects received no child anxiety disorder diagnosis using the parent structured interview and five of six received none using the child structured interview. After the follow-up period, four of six children received no child anxiety disorder diagnosis using the parent interview and four of six received none using the child form.

DISCUSSION

The present results provide support for the effectiveness of a family-based cognitive-behavioral therapy in treating children with a diagnosed anxiety disorder. Multimethod assessment indicted gains from pre- to posttreatment on parent-report, child-report, teacher-report, and independent clinicians' diagnostic ratings for four of six subjects. Further, the remaining two subjects showed improvement on most measures. Changes in parent, child, teacher, and clinician ratings after intervention contrast with the general stability of the children's difficulties and the fact that diagnostic interviews by independent clinicians found diagnoses to remain unchanged over baseline periods.

Presenting symptoms had been a clinically significant problem for each of the six subjects for at least 1 year prior to intake. The results represent clinically significant changes in the subjects' functioning, for many target complaints represented seriously dysfunctional behavior. For example, one subject had attended school only a few weeks during a 2-year period, and for four of six participants, school performance was seriously affected by separation fears, anxiety in social interactions, or evaluation fears. In addition, at least one target complaint for each child represented a situation in which anxiety seriously interfered with normal peer functioning.

Pretreatment results indicated the stability of the children's comorbid diagnoses and depressive symptoms during the baseline period. Comorbid conditions are typical in childhood anxiety disorders (Kendall & Brady, 1995) and comorbidity can contribute to limited treatment effectiveness. Results at posttreatment were particularly strong given the fact that intake assessment found that (a) half of the subjects had comorbid DSM-III-R diagnoses (e.g., attention deficit hyperactivity disorder and oppositional defiant disorder), (b) parents of all subjects reported clinically significant child depressive symptoms on the CBCL, (c) independent teacher ratings using the TRF showed clinically significant depression and anxiety for five of six subjects, and (d) four of six children self-reported a clinical level of interfering depressive symptoms.

Changes produced in subjects' ability to cope in individualized target anxiety-provoking situations are important for considering clinical efficacy. For example, one subject's CQ-C scores for the question, "When I have to go to school, I am able to help myself feel better," represented a change from school refusal for all but a few weeks of the prior 2 school years to daily attendance (including walking to and from school independently) at the end of treatment and follow-up.

Treatment gains were generally maintained during and at the end of the 4-month follow-up period, except for the one child who experienced parental separation and family violence during follow-up. The impact of significant life events suggests the need for and desirability of a parent-report measure assessing the stability of external stressors that might affect the child's symptoms and ability to cope.

There were advantages and disadvantages in the use of a multiple-baseline design in this study of family cognitive-behavioral therapy. The design helped to assure that observed effects were associated with intervention but, nevertheless, there are important limitations in using a multiple-baseline design to understand and treat complex disorders. For example, the design does not permit examination of differential outcomes for different diagnoses, relationships between comorbid conditions and outcome (Kendall, Kortlander, Chansky, & Brady, 1992), or for an unraveling of the multifaceted treatment. Also, the relative effectiveness of this family-based treatment remains unexamined (see also Barrett et al., 1996). A controlled group comparison study comparing a family therapy to a waitlist condition and/or to a child therapy is needed.

It is possible that improvement in internalizing symptoms and anxious behavior occurred in family members who, unlike the child, were *not* the target of treatment. Clinical observation and family members' anecdotal reports suggest that such unassessed positive outcomes occurred. At least one parent in four out of the six families reported serious personal problems with internalizing symptoms dating to their own childhood or adolescence: Each parent stated that he or she had benefitted in terms of a reduction of his or her symptoms from participation in the child-based treatment. Accordingly, we are currently conducting assessments of family interaction patterns and parent attitudes and psychopathology to better evaluate the possibility that family therapy for a child disorder has beneficial "spillover" effects (Kendall & Flannery-Schroeder, in press) on similarly distressed parents.

Based on the current research, family member's reports of functioning suggests that present family report measures may be insufficient to address the nature of family environments associated with child anxiety disorders. Perhaps, despite the greater procedural difficulties, it is valuable to include clinician ratings and/or behavioral observations of family interactions in future study of the families of children with anxiety disorders. It might also be that relevant interactions in families of children with anxiety disorders may be more subtle, have a lower base rate, and be more affected by social desirability than behaviors in families with acting out children—and thus, less open to reliable observation. For the anxiety disorders, it may be most useful to focus on family interactions during treatment. Clinical observation during this family treatment outcome study and other data (e.g., Siqueland et al., 1996) suggest that ratings of affective expression, avoidance of conflict, and expression of dependency needs may be particularly important.

The present study focused on changes in individual cases. Nevertheless, it is important to consider consistencies across subjects. Observations across families suggests the importance of social negotiation processes in treatment outcome. Also, there were changes seen across families: increased parental tolerance of child negative affect (e.g., sadness, anxiety, and frustration); the importance of humor in reducing demands for perfect performance; the normalization within the family of anxious experience; and the redefinition of previously frustrating situations to be avoided as problems to be solved. Future applications will benefit from an incorporation of these elements in both child- and family-focused interventions.

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